Aniruddha Pandey

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Chat with this resume at self-aware-resume.vercel.app. I'm in my 4th semester of master's at Stevens Institute of Technology. I'm fortunate to be advised by Dr. Nikhil Muralidhar. My current research interests are symbolic regression, LLM pretraining, reinforcement learning, and agent memory architectures.

Education

Jan 2025 – Dec 2026 (Expected)
M.S. in Computer Science - Stevens Institute of Technology, Hoboken, NJ

Coursework: Statistical Machine Learning, Deep Learning, NLP, Scientific Discovery in ML, Advanced Algorithms. **Volunteer**: Next-Gen Healthcare Innovators Symposium 2025 (panel curation, speaker coordination, event logistics, student research pitches).

Sep 2021
B.Tech in Computer Science and Engineering - Lovely Professional University, India

Graduate Research

Self-Aware Resume: 83M hybrid model (attention and **Mamba-3** in parallel) with a copy head that quotes from the document in context; pretrained on 100B tokens on 4×**H100**, then tuned with **SFT** and **GRPO**. Scores 0.80 on resume QA vs 0.15 for **Falcon-H1-Tiny-90M** and **SmolLM2-135M**; runs on a single CPU core in under 1 GB of memory (0.14 s per token) via **ONNX Runtime**.

Tri-Mem: Modality-matched memory for multi-agent systems; semantic memory (sparse-attention KV chunks), episodic memory (visual compression), and declarative memory (vector RAG) with entropy-driven routing. Cuts token cost by 84% on a custom NovaCorp AuditBench and reaches 81.4% on **LongMemEval**. *(Presented at the ICNS Symposium, Spring 2026)*

NeuralSINDy: Extending **SINDy** with neural function approximators to discover governing equations from noisy, high-dimensional dynamical systems data. *(In preparation)*

Work Experience

Jun 2026 – Aug 2026
AI Engineer Intern - Scholera, Hoboken, NJ (Hybrid)

Built an adaptive quiz engine on a 3PL Item Response Theory (**IRT**) model, with **EAP** ability estimation and Fisher Information item selection; developed a multi-stage RAG pipeline with Cross-Encoder reranking, citation checks, and out-of-corpus refusals, evaluated with **RAGAS** (MRR, Recall@K, faithfulness).

Feb 2024 – Nov 2024
Senior Frontend Engineer - TaxTaker, Austin, TX (Remote)

Designed an encrypted **localStorage** library for secure client-side data handling.

Oct 2022 – Sep 2023
Data Visualization Engineer - Spontivly, St. Petersburg, FL (Remote)

Built interactive dashboards for enterprise clients including **Balenciaga** and the **NFL**.

Sep 2021 – Oct 2022
Junior Frontend Engineer - Astro Malaysia, Kuala Lumpur (Remote)

Built an interactive tournament medal table for Commonwealth Games 2022 on stadiumastro.com.

Skills

Math: linear algebra, information theory, bayesian inference, probability theory, convex/stochastic optimization

Modeling: transformer architectures (encoder/decoder, sparse attention, KV caching), tokenizer design (BPE), distributed training (DDP, torch.compile, bf16), quantization, mechanistic interpretability

Systems: Python, PyTorch, vLLM, W&B, Slurm

Awards

RAISE-26 Finalist | 2022 SIH Judge | 2021 Microsoft Imagine Cup World Finals Runner-Up | 2020 HackMIT Health Tech Winner | 2019 SAP Analytics Cloud Winner